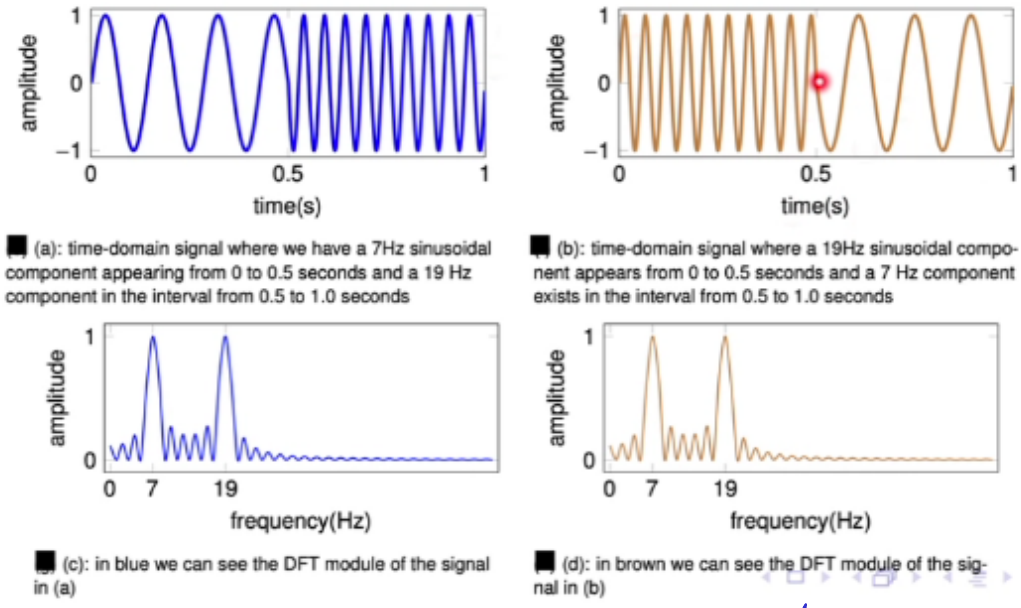
27/03/26

→ Fourier analysis → Time domain to frequency domain
↳ it has some drawbacks.



(a): time-domain signal where we have a 7Hz sinusoidal component appearing from 0 to 0.5 seconds and a 19 Hz component in the interval from 0.5 to 1.0 seconds

(b): time-domain signal where a 19Hz sinusoidal component appears from 0 to 0.5 seconds and a 7 Hz component exists in the interval from 0.5 to 1.0 seconds

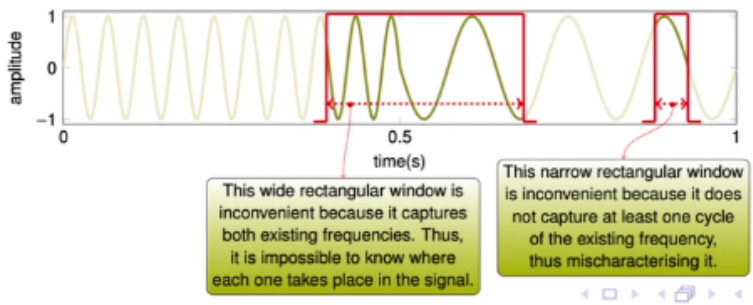
(c): in blue we can see the DFT module of the signal in (a)

(d): in brown we can see the DFT module of the signal in (b)

DFT for them. They are the same even though they are reversed. It wouldn't allow you to understand their position within the signal (time-support of frequencies) where the frequencies are located

→ To improve that we could use STFT (Short time Fourier transform). With it you make analysis of parts of your signal and check the light peak of it. It's a sliding window through time

- Particularly, narrow windows may mischaracterize low frequency components because they need sufficient time to complete at least one cycle. Conversely, wide windows may capture more than one frequency component occurring separately in the corresponding time interval, forbidding the intended localization, i.e., the condition of stationarity may be violated. In other words and as stated by Heisenberg's uncertainty principle, high time-resolution implies low frequency-resolution and vice-versa. This time-frequency tradeoff is illustrated in the next Figure.



↳ This approach helps, but there's another issue: the window size. If this is too wide, we risk in the analysis we wouldn't find the position. On the other hand, if the window is too short we'll have too much work and the wave might be mischaracterized.

↳ The new problem can also fit in Heisenberg's uncertainty principle - as higher the resolution in time, the lowest it is in frequency

→ FFT is good if you don't care about time support of frequencies

→ For voice recognition can used to have the moments, robust frequencies happen → for RNN as well

→ Wavelet can help finding the time support of frequency for images

→ There're two types → continuous and discrete. Both convert wave form time/space to domain time-frequency & even space-frequency

→ At the beginning wavelet was a tool for pure math because there were no possible applications at that time

→ Sin and cos are basis for Fourier. They have long infinite extent. These functions are the basis to convert input signals

→ Wavelets adapt time localized functions. It catches patterns of a signal. It's like the idea of windowing but with auto modified size

→ wavelet transform → mathematical tool

→ wavelet → a small wave

→ WT use wavelets and wavelet functions

→ CWTs → use just one small wave called wavelet function or mother wavelet. Map to time-frequency (joint) the signal can be dilated or shifted

in general they are not used

→ DWTs → use two wavelets. Known as scaling function or father wavelet. Also use dilating and shifting on the signal (an orthogonal)

→ DWT and CWT are used to analyze continuous signals

→ For discrete signals we use DTWT (Discrete-Time Wavelet Transform). Sometimes DWT is used to refer to DTWT. It's used both for time (1D) and space (2D)

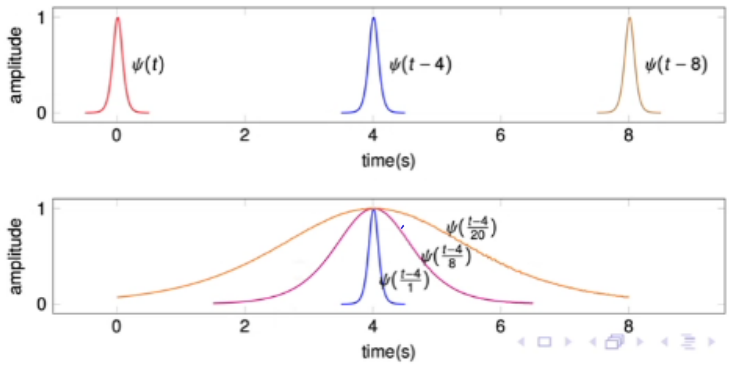
→ CWT working

- - - Wavelet Transforms - - -

- CWT working: Let $\psi(t)$ be the wavelet function of time t, regardless of its shape. Then, $\psi(t-b)$ for $b \in \Re$ is its t-seconds shifted version. Equivalently, $\psi(t-b)$ is the sliding window for which placement depends on b, as shown in the first Figure below. Accordingly, we can insert the parameter $a \in \Re$ so that the function becomes $\psi(\frac{t-b}{a})$. As a varies, the wavelet function is dilated, i.e., the window length is modified, as we can observe in the second Figure. Thus, $\psi(\frac{t-b}{a})$ is the sole function intelligently controlling the entire process.

a is the size of the window

dilating



→ STFT → makes a sharp break, but WT doesn't

This is the basis of CWT, you use $\psi(t)$ and integrate with the signal to get a result

- - - Wavelet Transforms - - -

- Grouping together all the concepts exposed up to this point, we can formally define the CWT as

$$C(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t)\psi\left(\frac{t-b}{a}\right) \, dt$$

← it's a convolution

normalization

← it's a bandpass filter

where $x(t)$ is the input signal under analysis and $\frac{1}{\sqrt{a}}$ is simply an energy normalization factor. CWT working becomes clearer based on the Equation above, i.e., $C(a,b)$ correlates $x(t)$ with $\psi(\frac{t-b}{a})$ over time and according to the scale, quantifying the resulting likelihoods, as shown in the example signal in the first Figure on the next slide. Colors and their intensities playing the role of a third axis in the graphical representation in the second Figure, are used to represent the respective correlation amplitudes. Thus, $C(a,b)$ converts the input signal from the time to the time-frequency domain, where the latter is formed by a set of likelihood amplitudes.

DTWT is low/high pass filtering

you're filtering low frequencies

→ Depending on the format and size of the window it can have different behaviour in frequency

→ CWT can't be done $-\infty \to \infty$, so we need DWT

→ DWT is the same as CWT, but instead of infinite window/sliding, we could do it in a different way. However, even in this new setup, we would have problems. We need it to cover over the 0 hz, but the function (CWT) is a bandpass filter, that cannot reach it. To fix it we use another function (scaling function) that has a low pass filter

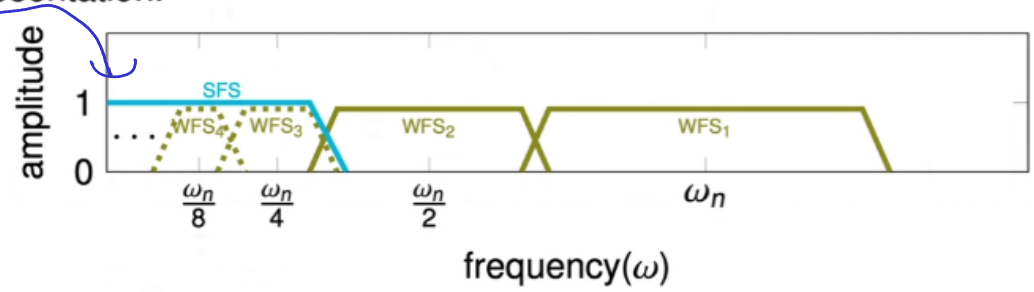
(in an both together). scaling $\phi(t)$

$\phi(t)$ must be orthogonal of $\psi(t)$

- - - Wavelet Transforms - - -

- Equivalently, DWTs allow us to represent a signal in terms of shifted and dilated versions of $\psi(t)$ complemented with the contribution from $\phi(t)$. Thus, the scaling and wavelet functions, also known as *father wavelet* and *mother wavelet*, respectively, form a basis for signal representation.

scaling function



- In the Figure, we can see, in cyan, the scaling function spectrum (SFS) overcoming the infinite number of stretched wavelet functions that would be required to reach the frequency 0Hz. In olive color, we can see four example wavelet function spectrums (WFSs).

→ DTWT → given a signal of M-sample $s[\cdot]$, DTWT is another ~~other~~ discrete-time signal $y[\cdot]$ with the same length

→ the transformed signal $y[\cdot]$ gives you the frequencies and their time-support

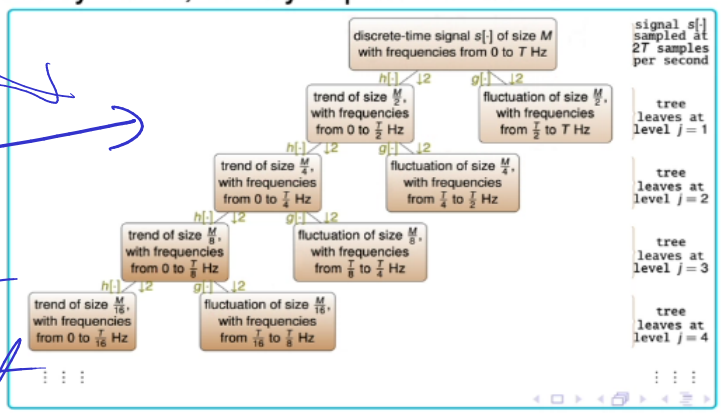
→ Alfred Haar is the father of WT (1910)

→ his idea was, if Fourier had used as and in which an infinite, what filter can happen if we used finite function? He used linear algebra function, this combination of two orthogonal function

- - - Wavelet Transforms - - -

- For decades, the linear algebra-based approach and the filter-based approach were considered separately by mathematicians and applied professionals, respectively. Later, the connection between both approaches was established... Particularly, the filter-based approach, shown in the Figure below, characterizes the *multiresolution analysis* proposed by Mallat, usually depicted as a tree.

It's a condition

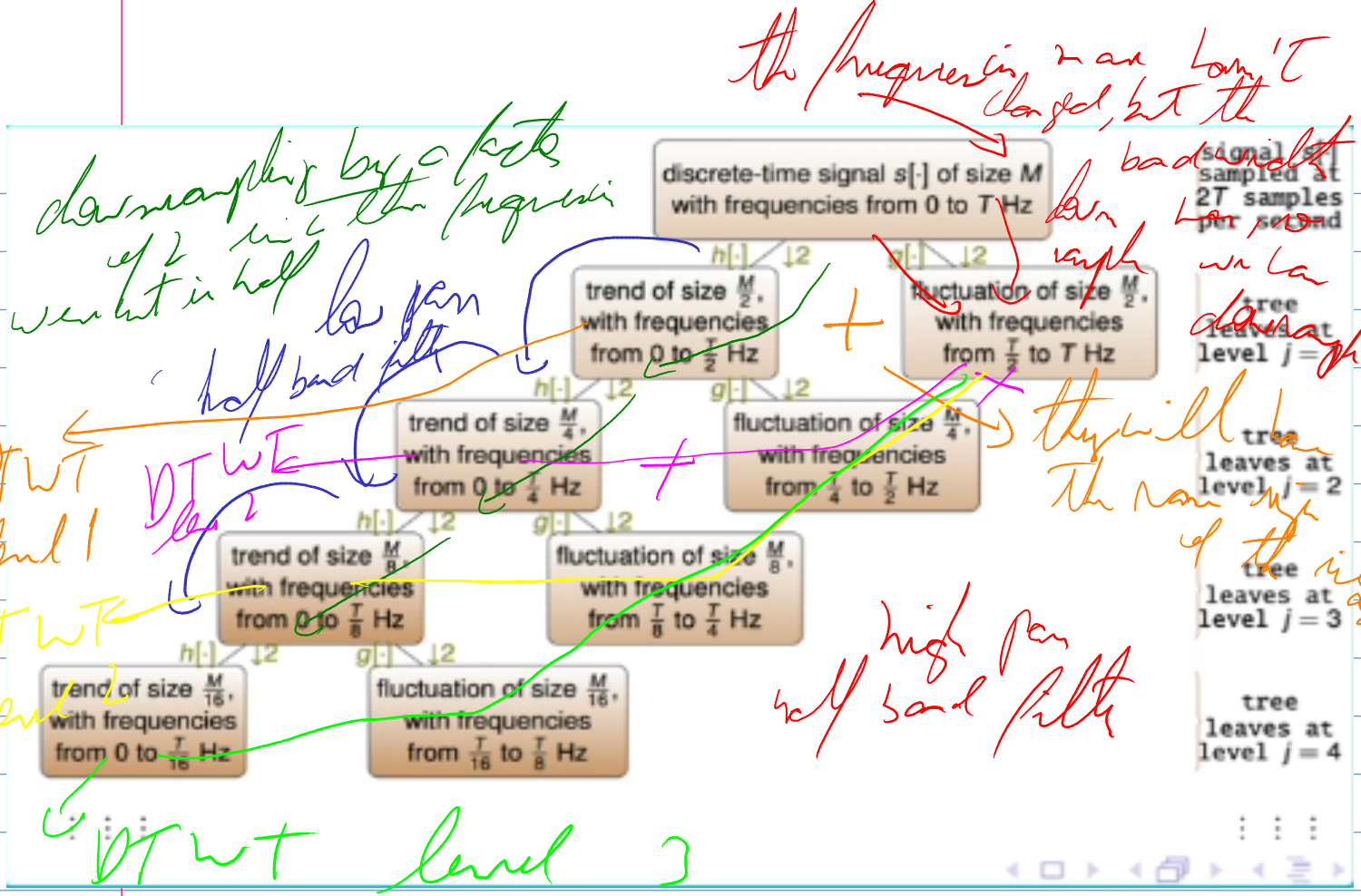


use of filtering that can retrieve the frequencies and time-support

Wavelet defined as DTWT

this can be random (have an inverse function)

the inverse wavelet function as proposed by Mallat. Since then it has been called wavelets



→ The sampling theorem doesn't say that the T/2 frequency exist, but it tells that it can accept

→ h → filter low pass filter with a half band cut

→ g → high pass half band filter

→ the bands of each level when concatenated with the g[·] + down sample result of previous level, keep the signal size and design a DTWT level of filtering